

Cactus pests

Find the pest, inspect the whole plant, and choose a response that matches the evidence.

Start with an insect, not just a mark

Separate the affected plant and inspect its tip, areoles, underside, pot rim and drainage holes with magnification. Photograph what moves. Natural wool, mineral deposits and old scars can resemble pests.

Mealybugs

Look for a segmented insect beneath cottony wax. Check sheltered crevices and nearby plants.

Root mealybugs

If growth declines without a visible above-ground cause, inspect roots and pot walls for wax and insects.

Scale

Look beneath a small cover for the insect. Dead covers can remain after treatment; monitor for living insects and new crawlers.

Mites

Stippling or bronzing needs a magnified inspection. Webbing can help, but its absence does not rule out mites.

Thrips

Inspect distorted tips and flowers for slender moving insects and dark fecal spots; scarring alone is inconclusive.

Fungus gnats

Monitor adults near damp media. Larvae can damage roots, especially in propagation trays; they are not merely a cosmetic nuisance.

UC IPM · Mealybugs · UC IPM · Houseplant problems

A measured response

- 1. Contain and document.** Keep suspect plants and their runoff away from the collection. Record the date, plant and evidence of live pests.
- 2. Remove accessible insects.** For a small infestation, use a gentle mechanical method that does not tear the epidermis. Reinspect hiding places rather than treating once and assuming success.
- 3. Use a matched treatment.** UC IPM includes a 70%-or-less isopropyl dab for houseplant mealybugs, with a small-area phytotoxicity test first. This is not a guarantee that alcohol is harmless to every cactus or suitable as a whole-plant spray.
- 4. Escalate deliberately.** For recurring infestations, confirm the pest and choose a locally permitted product labelled for the plant, setting and application method. Follow its interval, protective-equipment and pollinator restrictions. Do not improvise a root soak from a spray label.

5. **Verify control.** Inspect new growth and hidden sites over successive checks. A clean-looking day is not proof that eggs or underground pests are gone.

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What the cactus research actually shows

Cactus host studied A laboratory study of *Diaspis echinocacti* on *Opuntia stricta* found development and survival changed with temperature. That helps explain why inspection frequency matters in a warm collection; it is not a recipe for heating plants to kill pests.

Albuquerque Junior et al. · Cactus scale on *Opuntia stricta* (2023)

Two useful findings, with limits

Other hosts studied A scale-control experiment found treatment outcomes varied by scale species and life stage. The paper did not test every cactus-scale combination.

A root-mealybug trial used *Sedum*. It provides treatment leads, not a cactus root-drench rate or proof of safety on *Lophophora*.

Quesada & Sadof · Scale life stage and treatment (2017) · Gill & Kunkel · Root mealybug management on *Sedum* (2021)

Why a pest name can be misleading

Taxonomic research Work on the *Hypogeococcus pungens* complex shows why host association and genetic evidence matter. A familiar pest label may cover biologically different lineages. Do not transfer a biocontrol result to an unidentified insect in your collection.

Poveda-Martínez et al. · *Hypogeococcus* species complex (2019)

Keep growing

For collapsing tissue or spots without an insect, continue to Cactus health. See GeeBee: practice and evidence for how grower observations fit into the research. Practice observation and response in Gritty Mix. Its events and recovery speeds are simplified game mechanics, not treatment trials.

Interactive inspection assistant and current source links: revolutionarydesigns.io/cactus/guides/pests.html